

Peach LCD – ESP32-S3-DevKitC-1 Project Pinout

TOP VIEW (USB-C connector at bottom)

Function	GPI0		GPI0	Function
GND	GND	1	1	GND
3.3 V	3V3	2	2	5 V (VBUS)
Reset	[RST]	3	3	43 USB-Serial TXD0 (debug)
MCP23017 button IRQ	4	4	4	44 USB-Serial RXD0 (debug)
Touch INT (nc)	5	5	5	1 I ² C SCL (touch+buttons)
LCD D0	6	6	6	2 I ² C SDA (touch+buttons)
LCD D1	7	7	7	42 SD MISO
Radio UART2 TX	15	8	8	41 SD SCLK
Radio UART2 RX	16	9	9	40 SD MOSI
LCD VSYNC	17	10	10	39 LCD DISP (display on/off)
LCD D2	18	11	11	38 LCD Backlight PWM
LCD D3	8	12	12	37 free
free	3	13	13	36 free (input-only)
free	46	14	14	35 free (input-only)
LCD PCLK	9	15	15	0 BOOT button (strapping)
LCD D4	10	16	16	45 free (strapping)
LCD D5	11	17	17	48 onboard RGB LED (nc)
LCD D6	12	18	18	47 SD CS
LCD D7	13	19	19	21 LCD DE (data enable)
LCD HSYNC	14	20	20	20 USB D+ (native USB, nc)
LCD DE / free	21	21	19	19 USB D- (native USB, nc)
J3 (left header)		[RST btn] [BOOT btn] [USB-C]		J2 (right header)

nc = not connected in this project

BOTTOM VIEW (board flipped left-right, USB-C at bottom)

GPI0	Function	GPI0	Function
GND	GND	GND	GND
5V	3V3	3V3	3V3
43	[RST] Reset	[RST]	Reset
44	4 MCP23017 button IRQ	4	MCP23017 button IRQ
1	5 Touch INT (nc)	5	Touch INT (nc)
2	6 LCD D0	6	LCD D0
42	7 LCD D1	7	LCD D1
41	15 Radio UART2 TX	15	Radio UART2 TX
40	16 Radio UART2 RX	16	Radio UART2 RX
39	17 LCD VSYNC	17	LCD VSYNC
38	18 LCD D2	18	LCD D2
37	8 LCD D3	8	LCD D3
36	3 free	3	free
35	46 free	46	free
0	9 LCD PCLK	9	LCD PCLK
45	10 LCD D4	10	LCD D4
48	11 LCD D5	11	LCD D5
47	12 LCD D6	12	LCD D6
21	13 LCD D7	13	LCD D7
20	14 LCD HSYNC	14	LCD HSYNC
19	21 LCD DE	21	LCD DE

(J2 now left) (J3 now right) – pin order top-to-bottom unchanged

USB-Serial TXD0=43 RXD0=44 | SD MISO=42 SCLK=41 MOSI=40 CS=47
I²C SCL=1 SDA=2 | LCD BL=38 DISP=39 | BOOT=0 USB-D+=20 USB-D-=19

Peach LCD – ESP32-S3 Functional Pin Summary

LCD Panel JD9165A (8-bit parallel RGB, DE-mode)

Data bus	D0 = I06	D1 = I07	D2 = I018	D3 = I08
	D4 = I010	D5 = I011	D6 = I012	D7 = I013
Control	PCLK = I09	HSYNC = I014	VSYNC = I017	
	DE = I021	DISP = I039	BL = I038	(backlight PWM)

Touch HY4633 / FT6206-compatible (I²C addr 0x38) – shared bus

SDA = I02 SCL = I01 INT = I05 (pin defined, not wired in lcd.c)

Buttons MCP23017 GPIO expander (I²C addr 0x20) – same bus

SDA = I02 SCL = I01 INT = I04

SD Card (SPI)

MISO = I042 MOSI = I040 SCLK = I041 CS = I047

Radio Link (UART2)

TX = I015 RX = I016

Fixed / Board

USB-Serial bridge	TXD0 = I043	RXD0 = I044	(programming / debug)
Native USB	D- = I019	D+ = I020	(not used by firmware)
BOOT button	I00	(strapping pin – hold low at reset for flash mode)	
Onboard RGB LED	I048	(not used by project)	

Free GPIOs

I03, I022 – I037 (excluding I038, I039), I045, I046

Note: I035, I036 are input-only on the DevKitC-1 (no pull-up/pull-down)

Pin assignments read from source files:

components/LCD/include/lcd.h	– LCD data/control, touch I ² C, MCP23017 INT
main/app_main.h	– SD card SPI, radio UART2
Hardware/UartHW.h	– UART2 port number